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A Tangled Web: Should Online Review Portals Display Fraudulent Reviews?

Uttara M. Ananthakrishnan,^a Beibei Li,^b Michael D. Smith^b

^a Foster School of Business, University of Washington, Seattle, Washington 98195; ^b Heinz College of Information Systems and Public Policy, Carnegie Mellon University, Pittsburgh, Pennsylvania 15213

Contact: uttara@uw.edu,  <https://orcid.org/0000-0002-9341-1807> (UMA); beibeili@andrew.cmu.edu,

 <https://orcid.org/0000-0001-5466-7925> (BL); mds@cmu.edu,  <https://orcid.org/0000-0001-6844-7923> (MDS)

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Abstract. The growing interest in online product reviews for legitimate promotion has been accompanied by an increase in fraudulent reviews. However, beyond algorithms for initial fraud detection, little is known about what review portals should do with fraudulent reviews *after* detecting them. In this paper, we address this question by studying how consumers respond to potentially fraudulent reviews and how review portals can leverage this knowledge to design better fraud management policies. To do this, we combine theoretical development from the trust literature with randomized experiments and statistical analysis using large-scale data from Yelp. We find that consumers tend to increase their trust in the information provided by review portals when the portal displays fraudulent reviews along with non-fraudulent reviews, as opposed to the common practice of censoring suspected fraudulent reviews. The impact of fraudulent reviews on consumers' decision-making process increases with the uncertainty in the initial evaluation of product quality. We also find that consumers do not effectively process the content of fraudulent reviews (negative or positive). This result furthers the case for a decision heuristic that will incorporate the motivational differences between positive fraudulent reviews and negative fraudulent reviews to effectively aid consumers' decision making.

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Keywords: fraudulent reviews • randomized experiment • user trust • online platform

1. Introduction

O, what a tangled web we weave,
When first we practise to deceive! —Sir Walter Scott, 1808

With over half of the internet's population accessing social media sites regularly, online reviews have become a major source of product information for consumers. A 2016 survey by the Pew Research Center, for example, found that 82% of adults read online user reviews before making a purchase decision (Smith and Anderson 2016).

However, the increased importance of social media for (legitimate) promotion has led to increased incentives for, and an increased prevalence of, fraudulent reviews on major product-search websites. Fraudulent reviews occur when firms artificially inflate ratings of their own products or artificially lower the ratings of a competitor's product by generating fake reviews, either directly or through paid third parties. About 15%–30% of all online reviews are estimated to be fraudulent by various media (Streitfeld 2013, Bhattacharya 2017) and industry reports (Sussin and Thompson 2012), and the academic literature has examined the presence of such reviews on prominent

sites including Amazon (e.g., Feng et al. 2012), Expedia, TripAdvisor (e.g., Ott et al. 2011, Mayzlin et al. 2014), and Yelp (Luca and Zervas 2016).

This prevalence of fraudulent reviews is a significant threat for online review portals and product search engines given the potential for damage to consumer trust. For managers, addressing this threat involves main two steps: first, identifying which reviews are fraudulent, and second, determining what to do with potentially fraudulent reviews. There has been a great deal of work in the information systems (IS) and computer science literatures on identifying potentially fraudulent reviews.¹ Our paper represents the first research we are aware of to address the second question: What should firms do *after* identifying potentially fraudulent reviews?

This question not only is important to the academic literature, but also has the potential to greatly inform managerial practice. Today there is no consensus among practitioners regarding what to do with fraudulent reviews. Whereas many review/product portals such as Google Local² and Flipkart³ quietly delete fraudulent reviews, others, such as Amazon, publicly acknowledge censoring fake reviews,

including filing public lawsuits against firms and reviewers suspected of gaming the system. Other review portals, such as Yelp, go one step further by making the fraudulent reviews visible to the public with a notation that it is potentially fraudulent.⁴ There is, of course, a natural tension between these various approaches: Displaying fraudulent reviews aims to discourage fraud by reducing information asymmetry and increasing the reputation risks of being detected. However, this approach can be risky for online platforms because it may highlight the prevalence of fraud in the mind of consumers, thus reducing consumers' trust in all reviews.

Unfortunately, there is very little guidance in the academic literature regarding how firms should balance these two competing interests. Theory from the marketing and IS literatures addresses questions around the origin and the economic impact of fake reviews, but does not indicate how the presence of fraudulent reviews may signal the quality of information on the platform. More importantly, theory does not indicate how fraudulent reviews may influence product demand through its impact on consumer beliefs about the platform's quality. Our goal in this paper is to both address this gap in the academic literature and to inform the managerial understanding of online review dynamics. We do this by using randomized user experiments to address the following three questions:

1. What changes occur to consumer behavior when fraudulent review information is displayed?
2. How should a review portal display fraudulent review information to increase consumers' trust in the platform?
3. When do consumers use fraudulent review information to inform their choices?

Answers to these questions can inform fraud management policies by allowing firms to adopt strategies that consider both consumer behavior (i.e., consumers' cognitive load in evaluating fraudulent review information and how this information influences their trust in the platform) and seller behavior (i.e., what incentives sellers have to post fraudulent reviews in the first place).

Our empirical approach to answering these questions uses three randomized user experiments based on a restaurant review portal we designed and implemented. These experiments address the three questions identified above: Experiment I reveals *what* changes occur to consumer behavior when fraudulent review information is displayed. Experiment II explores *how* this information should be displayed to increase trust in the platform while improving information transparency and reducing cognitive burden on consumers. Experiment III studies *when* consumers use fraudulent review information to

inform their choices and highlights the need for simple mechanisms to display information.

Our goals and main findings are as follows. In our first randomized experiment, we study *what* changes occur when fraudulent review information is displayed in the most basic form. Specifically, fraudulent reviews are marked as such and are displayed alongside non-fraudulent reviews. In this experiment, we find that consumers expand their choice set and spend more time in the restaurant pages when fraudulent review information is displayed than they do in its absence. More importantly, we find that flagging fraudulent reviews and displaying them alongside non-fraudulent reviews causes consumers to choose restaurants that have lower historical fraudulent activity more than when fraudulent reviews are censored. Thus, our results from Experiment I demonstrate that there is a significant change in the way consumers behave and make choices when fraudulent review information is displayed. This leads us to the next question of designing a platform that consumers will trust without increasing their cognitive burden.

In this second randomized experiment, designed to understand *how* fraudulent review information should be displayed, we study *platform design* in the context of fraudulent reviews by analyzing the relationship that exists between information transparency, trust, and cognitive burden. More specifically, we address two important managerial questions: Will displaying fraudulent reviews improve the trust users place in the platform? What is the optimal way to increase trust without increasing the cognitive burden on the users? To do this, we use techniques from behavioral economics to quantify users' trust for the website through a betting experiment. This experiment shows that displaying fraudulent review information, and reducing the cognitive burden by providing a summary score of potentially fraudulent activities, significantly increases consumers' trust in the information provided by the review portal.

We complement the results of this experiment with a supplemental survey of consumers. This survey analyzes how consumers perceive the display of fraudulent review information. The results of this survey show that consumers perceive displaying fraudulent reviews and providing a summary score of fraudulent review activities as an effective deterrence mechanism to discourage sellers from writing fraudulent reviews. Specifically, 80% of the users in our survey agree that they trust a review platform more if it displays fraudulent review information because businesses are less likely to write fraud reviews on these platforms. Together, the results of our experiment and survey are consistent with the theoretical concept of *calculative*

trust, which predicts that consumers trust a platform more when they believe that it is in the self-interest of businesses not to engage in review fraud because of their fear of their fraudulent reviews being made public. In short, in Experiment II, we show users' trust in a platform is specifically linked to how information about fraudulent reviews is displayed, and that the optimal design is one that provides a strong disincentive for businesses to engage in dishonest activities and where this disincentive is clearly apparent to platform users.

Finally, in our third experiment, we analyze *when* consumers use fraudulent reviews and *if* the polarity of the fraudulent reviews influences consumers' decision making. These are particularly important questions because 85% of the users in our survey believe that they should have a choice in viewing truthful and fraudulent information and the platforms should leave the choice up to them to decide whether they use fraudulent review information in determining the quality of a business. Therefore, we study the conditions of initial product quality under which users opt in to view and utilize fraudulent review information and whether users are efficient in processing this fraudulent review information. We find that displaying fraudulent reviews is more influential on consumer decision making when consumers have a higher initial uncertainty of product quality. We also find that consumers are relatively inefficient in how they process the content of fraudulent reviews (negative or positive fraudulent reviews). This result extends the case for a simple platform design with a single decision heuristic that incorporates the motivational differences between positive fraudulent reviews and negative fraudulent reviews to effectively aid consumers' decision making.

This research topic is particularly timely given the presence of misinformation on technology platforms, the incentives of actors to exploit anonymity online to manipulate consumer beliefs, and the influence these actions have had on consumer trust. Specifically, a 2018 study conducted by the Pew Research Center (Smith 2018) found that 72% of Americans believe that technology firms can hardly ever be trusted to do the right thing, and a 2019 Pew Research Center study (Mitchell et al. 2019) found that 56% of respondents expect online misinformation to get worse in the next five years.

Our work informs this phenomenon by developing a consistent and a *proactive* managerial policy for online platforms to provide an unbiased estimate of how fair they think the user reviews are instead of *reactive* policies and sporadic legal actions by major technology platforms against businesses that are involved in writing fake reviews. As all retailers race to increase their sales online, they also increase their

efforts in garnering consumer reviews to make their e-commerce platforms valuable for their users. This has even led to a practice called "review syndication," where different retailers share reviews from the same users on different platforms to create and expand an alternative review ecosystem to the ones offered by major e-commerce platforms (Weise 2017). The explosion of user generated content in all e-commerce platforms has also led to increase in fake reviews on *most* platforms and not just on major players in the e-commerce industries (Morris 2019). Major efforts by retailers to create thriving review ecosystems on all platforms, increasing consumer awareness of the incidence of misinformation online, and the generally negative consumer sentiment toward technology firms create a need for academic guidance to inform managers on how to improve the public's trust on information provided by these platforms.

Our results suggest that platforms that identify fraudulent reviews should display these suspected fake reviews to their users to increase consumers' trust in the platform and to alleviate uncertainty. However, platforms should display fraud review information in form of a decision heuristic that will reduce consumers' cognitive burden in processing the fraud review information. We also find that consumers factor fraudulent reviews into their decision-making process, which can in turn be leveraged by platforms to penalize dishonest businesses and to protect fair but vulnerable businesses. Together, these results provide strong managerial and policy guidance to developing truthful, transparent, and accountable online ecosystems.

2. Related Literature

Our study addresses the issue of fraudulent reviews, and specifically what review portals should do after identifying fraudulent reviews.

The issue of fraudulent reviews came to national attention recently after the attorney general of New York fined 19 companies over \$350,000 for flooding Yelp, Google Local, and other review portals with fake reviews (New York State Office of the Attorney General 2013). Similarly, regulation agencies such as the Federal Trade Commission in the United States, the Competition and Markets Authority in the United Kingdom, and the Italian Competition Authority have either fined businesses that have employed fraudulent review practices (Helft 2010, William 2016, Federal Trade Commission 2017) or forced platforms such as Airbnb (Masoni 2014, Brignall 2017) to develop better protections against fraudulent reviews. Platforms themselves have also gone on the offensive, bringing legal action against businesses suspected of writing fraudulent reviews on Amazon (Plaugic 2015, Conger 2016).

Given the importance of online reviews to both commerce and policy, it is not surprising that there has been considerable interest in the economics, marketing, and information systems literatures in studying fraudulent reviews. Whereas the broad definition of fraudulent reviews can include reviews with factual inaccuracies about a business and exaggerations or misrepresentations of the experience from actual consumers, we study fraudulent reviews in the context used by papers in this literature—self-promotional positive fraudulent reviews and negative fraudulent reviews to malign competitors. These papers generally look at the economic aspects of review fraud manipulation that happen on review platforms such as Yelp, Expedia, and TripAdvisor. For example, Mayzlin et al. (2014) analyze fraudulent reviews in the context of hotels by comparing reviews in Expedia with TripAdvisor and find that the presence of a competing hotel and characteristics of the competing hotel and the hotel ownership (chain versus independent, size of ownership) determine the amount and nature of fraudulent review activity. Similarly, Luca and Zervas (2016) establish that independent restaurants with low reputations and high levels of competition are more likely to commit review fraud. Separately, Lappas et al. (2016) study the vulnerability of individual businesses to fraudulent reviews and show that 50 reviews are enough to gain visibility over competitors, and that both positive fraudulent reviews and negative fraudulent reviews are equally effective under different settings. With respect to consumer responses to fake reviews, Zhao et al. (2013) use simulations to show that consumers' uncertainty increases with the increased likelihood of encountering fake reviews. Together, these papers establish the antecedents of and motivations for businesses to commit fraud, in contrast to our research, which examines the optimal responses of businesses after fraud is detected.

Our research is also related to studies that establish the importance of the online review ecosystem in consumers' decision making, product evaluations, and purchase decisions. Prior work has found significant economic effects from online reviews in variety of settings, including the effect of reviews on product sales and purchase decisions (Chevalier and Mayzlin 2006), the effect of ratings on hotel bookings (Ghose et al. 2012), and the effect of online reviews on movie box office performance (Dellarocas et al. 2007, Chintagunta et al. 2010). Recognizing the importance of online reviews for consumer decision making, subsequent papers in the IS and marketing literatures have developed algorithmic techniques to mine and predict the usefulness and the subjectivity of online reviews based on various parameters, providing context to which reviews have a higher probability of

influencing a customer's purchase decision (Archak et al. 2011, Ghose et al. 2012).

Another stream of literature that is important to our study examines the theoretical underpinnings of trust. The importance of trust in online transactions has been underlined by different studies (Brynjolfsson and Smith 2000; Gefen 2000, 2002). One widely used model of trust in the existing literature is the theory of reasoned actions (TRA) developed by Fishbein and Ajzen (1977) and a parsimonious version of this model developed by Davis (1989). The trust literature has also defined several antecedents to the TRA—knowledge-, calculative-, institution-, personality-, and cognition-based trust (Gefen et al. 2003). Two trust-based antecedents or trust-building processes have direct theoretical implications in our work—calculative-based and institution-based trust. Shapiro et al. (1992) defines calculative-based trust as the belief that the other party will not deceive for the fear of getting caught, that is, this trust is deterrence based. We draw on calculative-based trust to explain our experimental results. We also invoke institution-based trust (McKnight et al. 2002, Pavlou and Gefen 2004) to provide a theoretical explanation for our proposed interventions to leverage identified fraudulent reviews as a structural assurance.

Additionally, from a policy perspective, our study is related to the recent media coverage of fraudulent reviews (Belton 2015, Dwoskin and Timberg 2018) and particularly the recent policy interest in holding platforms accountable for spreading misinformation on the internet. For example, a 2016 Pew Research Center study found that two-thirds of Americans believe that false information on the internet creates confusion about basic facts (Barthel et al. 2016). One notable study in this literature is that by Vosoughi et al. (2018), who study the diffusion rates of falsehoods using rumor cascades on Twitter.

From an algorithmic perspective, our research relates to the accuracy of state-of-the-art fraud detection algorithms presented in the computer science literature. For managers to be able to reliably respond to potentially fraudulent reviews, they must first have accurate fraud detection filters in place. Fortunately, the state of the art in detecting fraudulent reviews from the computer science literature has advanced rapidly in the last few years, such that managers can reliably infer fraudulent reviews from the predictions of available algorithms (Jindal et al. 2010, Lim et al. 2010, Wu et al. 2010). For example, Mukherjee et al. (2013) apply their fraud detection algorithms to Yelp's data and find them to be around 86% effective at identifying fraud based on the characteristics observed by the researchers. More importantly, the authors note that "Yelp can employ many other behavioral features extracted from its internal data,

e.g., features based on IP addresses, geo-locations, network/session logs, mouse gestures, click behaviors, social network interactions (friend and fan relations) of reviewers at its website, etc.” (Wang 2010) to improve the accuracy of their internal fraud predictions. As such, the 86% accuracy observed by the authors should be seen as a strict lower bound on the true accuracy a platform could achieve with full information.

Indeed, platforms have large amounts of user data and invest significant engineering and manual effort in implementing fraud-fighting algorithms at scale to prevent abuse (Kantrowitz 2015, CBS News 2016, Parkin 2018). Reviewer data with anomalous posting patterns when combined with text-mining techniques (Jindal et al. 2010, Lim et al. 2010, Li et al. 2014) can effectively identify most common forms of review fraud, such as review farms or click farms (New York State Office of the Attorney General 2013) with high accuracy. In fact, in the papers by Li et al. (2017) and Xu and Zhang (2015), the authors use reviewer posting rates compared with other groups of reviewers to detect collusive fraud. Likewise, Gimenes et al. (2015) leverage the “lockstep” behavior or group interaction within a short time span to identify temporal fraudulent behavior with 95% accuracy.

Although platforms can use sophisticated algorithms from the computer science literature and proprietary data to identify fraudulent reviews, there is little guidance in the literature about what firms should do *after* identifying potentially fraudulent reviews. Reflecting this lack of consensus, some platforms like Yelp and TripAdvisor display fraudulent reviews and publicly identify firms that commit review fraud (Trujillo 2009, Van Buskirk 2010), whereas most review portals simply delete the identified fraudulent reviews without further comment. What is the ideal intervention after identification? This is the question we address in our research.

In addressing this question, our research makes two main contributions to the literature. First, we contribute to the stream of academic literature that studies fraudulent reviews. Whereas existing studies focus on identifying the antecedents of fraud and which businesses are likely to commit or to be vulnerable to fraud, we study this problem through a different lens: Should online platforms display these identified fraudulent reviews to increase the accountability of all the parties involved? This is a vitally important academic and managerial question given the importance of review platforms in online markets and the lack of consensus in managerial practice about how to respond to potentially fraudulent reviews. Second, our research responds to the call by Lazer et al. (2018) for interdisciplinary research on

redesigning information ecosystems that promote truth. Our research does this by recommending specific, managerially actionable interventions to create an information ecosystem that preserves consumer trust on the online platform.

3. Experiment I—The Impact of Displaying Fraudulent Reviews on User Behavior

The majority of review portals do not choose to display suspected fraudulent reviews once they are detected. These suspected fraudulent reviews are simply removed from the portal. For example, Google and Amazon use fraud detection algorithms to identify and remove potentially fraudulent reviews and do not provide information to consumers about which, or how many, reviews were removed as potentially fraudulent. On other end of the spectrum, Yelp uses a system that detects potentially fraudulent reviews and displays these reviews to its users as being potentially fraudulent.

In our first experiment, we want to understand *what* happens if a website decides to display fraudulent review information to its users in the most simple form, by showing fraudulent reviews along with non-fraudulent reviews: Are there any changes in the way users behave in the presence of these fraudulent reviews or are users indifferent to the presence of fraudulent reviews? Specifically, our experiment allows us to compare users’ behavior when fraudulent reviews are displayed in review portals with the currently popular *silent approach*, in which the portal does not mention the measures it takes against fraudulent reviews or even acknowledge the possibility of fraudulent reviews.

To achieve our goal, we designed and conducted a randomized experiment on Amazon Mechanical Turk (AMT) between February 21, 2014, and March 4, 2014, with 238 subjects. AMT has been widely used in behavioral experiments and has been accepted in the literature as a standard platform on which to conduct randomized online user experiments. In the IS literature, Ipeirotis (2010) and Ghose et al. (2014) have conducted detailed user studies and shown that the AMT population is generally representative of the overall U.S. internet population.

In this experiment, we designed and implemented an experimental restaurant review portal and a reservation system using real data, using a behavior tracking system to obtain key metrics such as time spent on each page, clicks, restaurant pages visited, and activities such as sorting by different parameters, and finally, the details of what restaurant was chosen by the subject. (An example screenshot is provided in Figure 4 of Online Appendix A.)

3.1. Randomized Experimental Design

We ran our experiment using a pure between-subject design. We consider two different scenarios, as described in Table 1. We implemented each of the scenarios as a real-life decision-making environment and assigned subjects randomly into one of the two scenarios.

In Scenario A, fraudulent reviews are marked clearly and are accessible to the user at the bottom of the review page. This scenario corresponds roughly to the fraud management strategy adopted by Yelp. Scenario B is the more widely deployed silent approach, where the fraudulent reviews are removed and are not displayed to the subjects. This scenario corresponds to the approach adopted by most review portals, including Google and Amazon. A pictorial representation of the differences between the scenarios is presented in Figure 3 in Online Appendix A.

The website interface began with instructions about the experiment followed by a main page. Then, each user was randomly assigned restaurants from any one of the five cities on our experimental website.⁵ The main page contained a list of 50 restaurants and a short summary of each restaurant's ratings, price, and total number of reviews. The order in which the restaurants were displayed was random for each user.⁶ This page also had options to sort by price and sort by rating. Once the subject clicked on a restaurant, they were taken to the restaurant's landing page, where they were shown a detailed description of the restaurant and reviews. The restaurant pages changed slightly in design, depending on the scenario the user was assigned to. Each user was exposed to only one scenario at a time. The user could go back to the list of the restaurants after browsing through each restaurant just like they would do in a real-life restaurant review platform.

Balance test results to ensure randomization are provided in Section A.1 of Online Appendix A.

3.1.1. Incentive-Based Quality Control. Paying attention to fraudulent reviews is a second level of investigation that happens in real settings when there is a considerable investment at stake. We tackled this issue by informing the subjects that they would be paid a bonus of 10 times their participation wage if the

restaurant they chose was among the top three restaurants judged by an internal panel unknown to the subjects. This method of incentivizing subjects is a common practice in the behavioral economics literature.⁷ We also informed the subjects that the best way to get this bonus would be to pay attention to all restaurant information displayed to them, such as rating, reviews, and price, and to choose a restaurant they would likely choose in the real life. By making this choice incentive compatible, we ensured that the subjects internalized their stakes and were incentivized to optimize their decisions, as they would in a real-life scenario. We also told the subjects that all the information that they would need to make a good decision was present on the website and that their chances of winning increased with how informed they were about the restaurant and how much attention they paid to the evaluation process. From the list of restaurants, the users could browse through each restaurant by clicking on each restaurant page, where they could browse through the details of the restaurant, reviews, ratings, etc. From the restaurant page, the users could either choose the restaurant and exit the experiment or go back to the main page and click on other restaurants from the list of restaurants to pursue a different final choice.

3.2. Results

3.2.1. Impact of Displaying Fraudulent Reviews on User Search Behavior. First, we analyze the experimental results at the user level, i . To do so, we run a simple ordinary least squares regression with number of clicks and total time spent by each user across all the restaurant pages as the dependent variables (DVs). We provide the summary statistics for the dependent variables in Table 2. We use demographic controls from our survey including age, income, internet usage, and review portal usage in the regression. We also control for the city that the user was exposed to in the experiment. These five variables are represented by the vector X_k below. We formally write the regression in Equation (1) and display the results in Table 3:

$$DV_i = \beta_0 + \beta \text{Scenario} + \sum_{k=1}^5 \alpha_{ik} X_k + \epsilon_i. \quad (1)$$

Table 1. Description of Experimental Scenarios

Scenario	Referred to as	Changes in restaurant page	Number of subjects
A	Fraud information displayed	Fraudulent reviews were displayed in a separate page, a link to which was present at the bottom of the restaurant page	112
B	Silent approach	This looked like a regular page from a restaurant review portal with no mention of fraud	120

Table 2. Summary Statistics for Experiment I

Analysis	Variable	Mean	Std. dev.	Min	Max
User level	Clicks	6.16	5.82	1	55
	Total time per user (mins)	3.75	4.74	0.27	48.36
Restaurant level	Total users	4.73	6.51	1	58
	Clicks	5.78	9.07	1	93
	Total time per restaurant (mins)	3.57	6.24	0	52.39

Note. Std. dev., Standard deviation.

Second, we analyze the data at the restaurant level, j , in each scenario. To do so, we aggregate the data at the restaurant level. We specify a model similar to Equation (1), but with restaurant-level characteristics m , including rating, number of reviews, category, city of the restaurant, and price, because the data are at the restaurant level. We run the regression using the activities on the individual restaurant page.⁸ Specifically, we employ clicks on each restaurant page, total time spent on each restaurant page across all users who visited the restaurant page (total time per restaurant, in minutes), and the number of visitors to each restaurant page (total users) as the dependent variables as shown in Equation (2). We present the results in Table 4.

$$DV_j = \beta_0 + \beta \text{Scenario} + \sum_{m=1}^4 \alpha_{jm} X_m + \epsilon_j. \quad (2)$$

Our results in Table 4 show that restaurants in Scenario A (fraud information displayed) had more visitors than those in Scenario B (silent approach). This finding indicates that consumers expand their choice set when fraudulent reviews are shown versus when they are censored. We also find that subjects spent more time in Scenario A both at the restaurant level and the user level. Note that in Scenario A, fraudulent reviews were grouped at the bottom of each page and were marked fraudulent, that is, they were not shown in-line with the non-fraudulent reviews. Because of this, it is likely that the time subjects spent on the platform was not used in evaluating whether each review was fraudulent because this information was readily available to them. This result suggests that users tend to make their choices more carefully, and use the information at hand more rigorously, when they are exposed to the suspected fraudulent information. This increase in search for

options can be explained by behavioral theory (Tversky and Shafir 1992), in that consumers are less likely to look for alternatives when they have a compelling reason to choose a particular option. However, they are likely to search for additional options when it is difficult to rationalize their choice. One can argue that in the absence of fraudulent review information, subjects were able to quickly rationalize a final decision, whereas they were less likely to do so in the presence of fraudulent reviews, where they looked for more alternatives. The increase in effort to rationalize is also explained by the increase in the number of clicks and the time on the web page.

More specifically, given an incentive-compatible task to find the best restaurant using the information presented to them, the subjects indeed behave differently in the two treatment scenarios. The restaurants in the experiment, on average, receive more visitors who spend more time on the page when fraudulent reviews are displayed than when these reviews are not displayed. This finding provides support to our result that by displaying fraudulent information, review portals may facilitate user engagement by attracting more users to participate in product search. However, it is possible that displaying the raw fraudulent review information might increase the number of attributes available to the consumers, which might force them to consider their options using lexicographic evaluation rules. This also shows that there is a nonnegligible cognitive cost associated with processing fraudulent reviews. This leads us to the question of how to display fraudulent reviews in a way that it improves the trust that the consumer places in the platform, without increasing the consumers' cognitive burden. We address this question in our second experiment.

Table 3. User-Level Regressions

	(1) Clicks	(2) Total time per user (mins)
Scenario A (Fraud information displayed)	1.801** (0.832)	1.457** (0.664)
N	232	232

Notes. The control variables are age, income, internet usage, review portal usage, and the city of the restaurant displayed. Robust standard errors are in parentheses.

** $p < 0.05$.

Table 4. Restaurant-Level Regressions

	(1) Total users	(2) Clicks	(3) Total time per restaurant (mins)
Scenario A (fraud information displayed)	1.628** (0.748)	2.570** (1.067)	1.614** (0.726)
N	256	256	256

Notes. The control variables are rating, number of reviews, category, city, and price of the restaurant displayed. Robust standard errors are in parentheses.

** $p < 0.05$.

3.2.2. Impact of Displaying Fraudulent Reviews on User Choice.

We also analyze the features of the restaurants that are ultimately chosen by the subjects in the different treatment conditions to see the effect of fraudulent reviews on consumers' choices. We model the choice of the subjects using logistic regression as follows:

$$\Pr(\text{Chosen})_{ij} = \beta_0 + \alpha_1 \text{FraudRatio}_i + \alpha_k X_{ki} + \gamma_u Y_{uj} + \epsilon_{ij}, \quad (3)$$

where $\Pr(\text{Chosen})_{ij}$ refers to the probability of restaurant i to be chosen by user j after the restaurant page is opened; α_1 is the coefficient of interest and represents the impact of the ratio of the number of fraudulent reviews to total number of reviews on the probability of a restaurant being chosen; X_{ki} is a vector of restaurant characteristics including rating, category, and price; and Y_{uj} is a vector of user-level characteristics including age, income, internet usage, and review portal usage. We run this model for Scenario A, where fraudulent reviews are displayed, and present the results in Table 5. Column (1) does not include user fixed effects, and column (2) includes user fixed effects to control for multiple observations from the same user.

We find that α_1 is significant and negative for Scenario A (where fraud review information is displayed). This result shows that subjects in Scenario A (fraud information displayed) chose restaurants with a significantly lower average fraud ratio (lower history of fraud). This result seems to indicate that, on average, subjects tend to associate the number of fraudulent reviews for a restaurant as a signal of poor quality.

3.3. Discussion

The main purpose of this experiment is to see what changes occur in consumer behavior when fraudulent reviews are (or are not) displayed to consumers on the restaurant pages. We measure these changes using key website-level metrics: clicks, time spent, number of product (restaurant) pages visited, and the activities on the restaurant pages themselves. We observe that there are significant changes in the subjects' behavior if they are shown fraudulent reviews along with the non-fraudulent reviews. To examine the underlying mechanism that drives these observed consumer behavioral changes, we conducted further experimental analyses, which we will discuss in the next two sections.

The results from this experiment also suggest that historical fraudulent review information plays a role in subjects' decision-making processes. This can be an effective tool for review portals to curb fraudulent behavior of dishonest businesses. If subjects associate fraud with poor quality and incorporate fraudulent review information in their decision making process, displaying this information can be a good way to introduce a penalty for getting caught and deter businesses from writing fraudulent reviews in the first place.

The metrics measured in Experiment I provide insight into the behavior of consumers when facing review portals with two different fraud management policies. Beyond the immediate user engagement and choice, a key factor that wins the loyalty of consumers and makes them use a review portal regularly is how this information is displayed to the consumers: What is the optimal platform design that can increase the

Table 5. Impact of Fraudulent Reviews on User Choice

	(1) Probability of chosen	(2) Probability of chosen
Ratio of fraudulent reviews	−4.702*** (2.770)	−7.976*** (2.256)
User fixed effects	No	Yes
Observations	659	594

Notes. The control variables are X_{ki} (restaurant characteristics—rating, category, and price) and Y_{uj} (user-level characteristics—age, income, internet usage, and review portal usage). Robust standard errors are in parentheses.

*** $p < 0.01$.

level of trust that consumers place in the information provided by the web portal? Will automatic filtering increase the trust that the consumers place in the platform? How does this connect to the trust formation process? We address these questions in a second randomized experiment.

4. Experiment II—Evaluating the Impact of Displaying Fraudulent Reviews on Trust

Building on the results from Experiment I, the primary objective of our second experiment is to understand *how* fraudulent review information should be presented to the consumers in a way that it increases users' trust in the information provided by the platform. Specifically, we study platform design and how the trust that consumers place in the review portal changes when the website attempts to display fraudulent review information. It is important to note that one major reason that most platforms do not display any information about fraudulent reviews is the *unknown* impact that fraudulent information might have on user trust toward *all* information on the platform.

Studying platform design in the context of fake information and the subsequent impact on consumers' trust is critical because of the growing awareness of misinformation on the internet and the growing mistrust of consumers on technology platforms. According to a 2018 Pew Research Center study, 72% of Americans believe that technology firms can hardly ever be trusted to do the right thing (Smith 2018). The problem of improving consumers' trust in online platforms is even more salient given the growing number of new entrants in e-commerce markets. These retailers, who have been steadily moving their business online, are also involved in building a robust review ecosystem to compete with other well-established players in this field and will potentially face a even bigger challenge in earning consumers' trust via reviews provided through online platforms.⁹

From a methodological perspective, one challenge in this process is how to properly measure consumer trust in the online marketplace. Most of the papers in the marketing literature use surveys to assess consumers' trust using yes/no or Likert-scale responses. Although this is an accepted technique, it is difficult to translate the amount of trust elicited through a survey into the actual amount of trust that a consumer would place in a web portal. Surveys are further limited in their ability to predict a consumer's trust in a system when there is no cost to making a bad decision.

This served as a motivation for us as we designed our second randomized experiment. We implemented a *betting experiment* to quantify the trust that a user places in the information provided by the review portal. This is a technique widely used in the behavioral

economics literature (Glaeser et al. 2000). Researchers in human/computer interaction (HCI) have also widely used coordination games/experiments (Riegelsberger et al. 2003a) or modified single-player versions of these games (Riegelsberger et al. 2003b) to measure trust in computer mediated communication. In our setting, this betting framework allows us to make our subjects financially invest in their decisions based on their perception of the veracity of information provided on the website. This provides a more accurate measure of what they would do in their real-life decision-making process when there is a monetary risk involved in an e-commerce transaction. Here the users of the portal decide whether to trust a product or a service based on the information present on the portal. The consumers are placing themselves at risk—in real life, of receiving bad services, and in our experiment, of a considerable loss of money—if they make a poor choice.

4.1. Randomized Experimental Design

The main goal of this randomized experiment is to study how managers should display fraudulent review information to increase users' trust in the information provided by the review portal. We focus on five major scenarios that are either popular in existing industry practice or theoretically beneficial to the review portal. In doing so, we aim to explore the most effective way to manage fraudulent review information.

In particular, first we considered the two major scenarios from the previous experiment, one displaying fraudulent reviews along with the non-fraudulent reviews and one displaying just the non-fraudulent reviews. Second, one major challenge consumers face today when making decisions on product search engines is high search cost, especially given an overwhelming amount of review content (Ghose et al. 2018). To help reduce users' cognitive burden of processing all the information, for each restaurant, we presented a *trust score* based on all its reviews to indicate the proportion of fraudulent reviews. We will elaborate the design of this trust score in the following subsection. Thus, we considered two additional scenarios in our experiment by adding this heuristic trust score to the above two scenarios. Finally, we added another control scenario where platforms can emphasize that the fraudulent-review filtering was done at the back end but does not provide any other information. Such a disclaimer to publicly acknowledge censorship is also widely adopted by several existing platforms (e.g., Amazon, TripAdvisor¹⁰). Overall, our goal is to understand and compare user preferences toward these five scenarios. A summary of descriptions of all the scenarios is presented in Table 6.¹¹ A pictorial representation of the differences between the scenarios is presented in Figure 1.

Table 6. Differences in Experimental Design

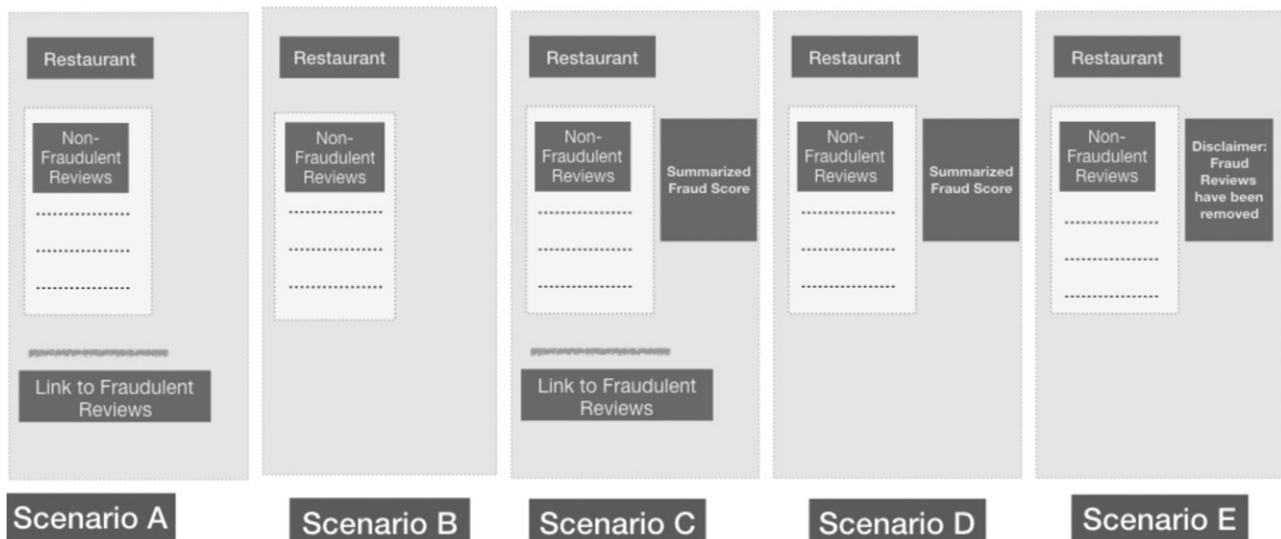
Scenario	Description
A—Fraud information displayed	Fraudulent reviews were linked to a separate page, and this link is present at the bottom of the restaurant page containing the non-fraudulent review.
B—Silent approach	Only non-fraudulent reviews are displayed. Fraudulent reviews are not displayed, nor is any indication given that any form of fraud review filtering was done at the back end.
C—Fraud and trust information displayed	This is the same as Scenario A but with a trust score. Fraudulent reviews are linked to a separate page, and this link is present at the bottom of the restaurant page containing the non-fraudulent reviews. A summary of fraud in the form of a score is provided.
D—Only trust score displayed	A summary of fraud in the form of a score is provided along with the non-fraudulent reviews.
E—Informed silence	Fraudulent reviews are not displayed, as in the silent approach, but the platform indicates it has taken care of the fraudulent review issues at the back end.

In our experiment, we built a website representing each of the five scenarios.¹² Note that all the scenarios displayed non-fraudulent reviews, but some also displayed fraudulent review information (i.e., all content of the fraudulent reviews or/and summarized trust score). Screenshots of the setup are provided in Section B.1 of Online Appendix B.

4.1.1. Implementation of Trust Game. We ran our experiment on AMT from September 7 to 10, 2018. Two hundred ninety-three subjects from the United States participated in the experiment. To understand user preferences toward each different scenario, in the main analysis, we applied a pairwise comparison approach (Ghose et al. 2012). The goal of this experimental design is to provide a point of reference to the subjects when they compare two designs in order to obtain a “ranked” preference, while avoiding overwhelming the subjects with too many choices for comparison at the same time. Note that as a robustness check, we have also implemented two alternative experimental designs to let subjects evaluate (1) all

scenarios at once or (2) only one randomly selected scenario. All tests demonstrate highly consistent results. We discuss the detailed results from our main pairwise design in this section. We also provide the details for our two alternative robustness tests in Section B.3 of Online Appendix B.

Each subject was randomly assigned a pair of websites out of the 10 possible combinations of five scenarios. Each of these pairs represent a pair of scenarios that we are interested in testing. We paid the subjects a small participating fee and then gave the subjects \$1 worth of virtual chips that they could use to bet on the site(s) they trusted more. In the instructions, we informed the subjects that these websites have previously participated in competitions, and through these competitions they have been categorized into how trustworthy they are in terms of the information that they provide to the users. We also informed the subjects that their bets should reflect the amount of trust they had in the site. They could bet on one or both sites. Not betting was not an option. In other words, they could divide their \$1 chips and

Figure 1. Differences Between the Experimental Scenarios

invest them among the two sites presented to them based on how much they trusted each site and they would be paid a bonus equivalent to the amount of chips they bet on a website (in addition to their participation wage) if that particular website had been judged to be more trustworthy based on the prior competitions. As explained above, we made trust incentive compatible by asking the subjects to monetarily invest on the platform based on the information provided to them.

Because trust takes different meanings in different contexts, in the instructions, we explicitly defined trust based on our context of a restaurant review portal as follows: “Think of trustworthiness in the real-world context—as a site you would believe gives you the best information and the one that you would want to use again and again if given the choice.”

Note that to avoid potential user confirmation bias, we referred to the fraudulent activity summary score displayed in Scenarios C and D as a “review quality score,” rather than using the word “trust” directly at a restaurant level. Sites displaying the review quality score had a short description of what it meant, along with the score. The score that was displayed to the subjects was just the proportion of nonfraud reviews scaled to a score from 1 to 5. The main purpose of this score was to see whether there was a major change in the amount of trust that the users placed on the review portal by displaying a decision heuristic. A short explanation of the review quality score was adapted from what Yelp uses under its nonrecommended review page. A screenshot of this score on the website is provided in Figures 8 and 9 in Online Appendix B.

As in Experiment I, we designed a system that tracked users’ activities throughout each website. In this way, we were able to confirm that all 293 subjects opened each website and the individual restaurant pages that contained the differences, before making their bets. As in the other experiments, we also included “attention check” questions in the experiment that could be answered only if the subjects opened the restaurant page. The subjects were told that their participation wage and bonus (if any from the betting) were subject to answering these questions correctly.

Kittur et al. (2008) show that such verifiable questions greatly improve worker attention and provide quality assurance. Table 7 provides the number of users for each of the 10 combinations of scenarios. To ensure randomization, we conducted balance tests for all the 10 possible combinations, and we present the results in Section B.2 of Online Appendix B.

4.2. Results

In our experiment, each subject split his or her bet of \$1 between two scenarios (i.e., two designs of the website presenting fraudulent review information in different ways) based on the amount of trust they placed in the scenarios. If the subjects perceived no differences in design, we would expect that the bets to be equally split between the two scenarios in order to maximize their winnings. In other words, a higher amount of bets placed on a scenario would indicate a higher level of trust and vice versa.

We analyzed the bets placed on each pair of the scenarios, presenting the results in Table 8. The first number in parentheses represents the average bets placed for the scenario corresponding to the column, and the second number in the parentheses represents the average bets placed for the scenario corresponding to the row. For example, the average bet for Scenario A (fraud information displayed) is 0.66 when it was presented in comparison with Scenario B (silent approach), whose average bet was 0.35. The asterisks represent the significance based on a *t*-test between the bets. Note that we are making pairwise comparisons between the elements in each cell, that is, we are not comparing the first element of one cell in Table 8 with the first element of another cell. For the discussion later, we will make use of the letters in the columns and the numbers in the rows to refer to cells corresponding to the pair under consideration.

Table 8 shows that there is a statistically significant difference between each of these pairs except between Scenarios D and E (cell D5), which is marginally significant. This implies that when presented with a pair of websites that were exactly the same except in the way fraud review information was presented, subjects evaluated each of these sites differently and

Table 7. Number of Subjects for Each Pairwise Comparison

	Fraud information displayed	Silent approach	Fraud and trust score displayed	Only trust score displayed	Informed silence
Fraud information displayed	X				
Silent approach	30	X			
Fraud and trust score displayed	25	26	X		
Only trust score displayed	30	31	28	X	
Informed silence	31	32	30	30	X

Table 8. Pairwise Comparisons of Bets

	Fraud information displayed (A)	Silent approach (B)	Fraud and trust score displayed (C)	Only trust score displayed (D)	Informed silence (E)
Fraud information displayed (1)	X				
Silent approach (2)	(0.66, 0.35)***	X			
Fraud and trust score displayed (3)	(0.33, 0.67)***	(0.36, 0.64)***	X		
Only trust score displayed (4)	(0.42, 0.58)**	(0.39, 0.61)***	(0.61, 0.39)**	X	
Informed silence (5)	(0.65, 0.35)***	(0.37, 0.63)***	(0.60, 0.40)***	(0.55, 0.44)*	X

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

placed bets that were significantly different between the groups. This shows that subjects appear to be cognizant of the changes in each of these sites. The result can be attributed to the reason that users considered some of the characteristics to be trustworthy, which in turn caused them to trust some sites more than others. We also note that if the subjects had been indifferent between the options, they would, on average, have bet an equal \$0.50 on both sites, which was not the case in our experiment. In particular, when looking into each pair and how the pairs rank with each other, we find the following:

1. We analyze the cases where Scenario A is compared with each of the other scenarios, B, C, D, and E. Scenario A (fraud information displayed) outperformed the two silent scenarios, Scenario B (silent approach) in cell A2 and Scenario E (informed silence) in cell A5, but it underperformed the two scenarios where the summary trust scores were displayed, Scenario C (fraud and trust score displayed), in cell A3 and Scenario D (only trust score displayed) in cell A4.

2. Now we compare the cases where Scenario C is compared with each of the other scenarios. The mean bet of Scenario C (fraud and trust score displayed) in cell A3 was consistently (and statistically significantly) higher than that all of the scenarios (as observed in cells A3, B3, C4, and C5).

3. Scenario D (only trust score displayed) performed better when compared with each of Scenario A (fraud information displayed) in cell A4 and Scenario B (silent approach) in cell B4. But Scenario D performed worse than Scenario C (fraud and trust score displayed), as represented in cell C4.

4. Scenario E (informed silence) performed better than Scenario B (silent approach), as seen in cell B5, but performs worse in the rest of the comparisons.

5. Finally and most notably, Scenario B (silent approach), which is the most popular approach that most review portals adopt, performed worse than all other designs.

The rankings obtained from these pairwise comparisons demonstrate that consumers place higher

levels of trust in designs that display fraudulent reviews, and moreover, a summarized version of the fraud review information in the form of a trust score. Our findings suggest that (i) users prefer a website design that can minimize information asymmetry (one that displays not only the legitimate reviews, but also the suspected fraudulent information), and (ii) users are likely to prefer a website design that can reduce their cognitive burden in processing fraud information in the form of a summary heuristic score, compared with parsing the raw fraud information from the fraudulent reviews. One such score would be the original rating (obtained by averaging over all non-fraudulent reviews), adjusted based on the number of fraudulent reviews and the probability of the reviews of being fraudulent. We present an approach to develop this score in Online Appendix D.

What is also noteworthy is that the currently popular silent approach in Scenario B was considered to be the *least* trustworthy among the scenarios. Review portals may not display fraudulent reviews out of fear from the potential loss in trust that consumers place in the website. However, our results suggests that by displaying fraudulent reviews and by decreasing the cognitive cost in processing fraudulent reviews, the website actually increases consumers' trust in the review portal.

4.3. Robustness Checks

We verified the robustness of these results using three other designs of the randomized experiment.

In the first robustness check, we conducted the experiment with 109 subjects and exposed them to all four scenarios at once using the same betting framework that we employed in the main experiment. In this test, each subject was asked to split \$4 worth of virtual chips among four designs of the website, represented by Scenarios A, B, C, and D. We find that the results are highly consistent with the users' ranking discussed in the main results, with users considering certain scenarios to be more trustworthy than others.

In the second robustness check, we exposed each subject to only one version of the website and asked them to rate the amount of trust on the features of the platform. Note that this experiment was not based on a betting framework, and we ran this experiment with 214 subjects. Again, this format of the experiment provided consistent results with the highest amount of trust placed in Scenario C, followed by Scenarios A and D.

In a third robustness check, we again exposed each subject to only one version of the website design and obtained his or her valuation through a similar betting framework. Specifically, we randomly assigned each subject into any one of the scenarios presented in Table 6 and asked the subject to provide valuation from \$0 to \$1,000 based on how trustworthy he or she thought the information on the website to be. Note that each subject saw only one design of the website. We ran this experiment with 298 subjects and found results consistent with those reported above.

Details about the experiment along with the results for these three robustness checks are presented in Section B.3 in Online Appendix B.

4.4. Discussion

Today, the business model of review portals is built around instilling consumer confidence on user-generated content and platforms. Platforms have invested in building fraud detection systems that operate at scale and with high degree of accuracy (Streitfeld 2011). Despite the growing instances of false information on the internet, the existing literature does not provide any practical recommendations on how to manage fraudulent reviews after they are identified or how to disincentivize businesses from committing fraud in the first place.

From Experiment II, it is evident that consumers' trust in the review portal increases when they observe a score that provides a summary of fraudulent activities and the quality of the customer reviews.

Our study suggests that displaying fraud reviews, and moreover, a review trust score, improves trust. We theorize that displaying fraudulent reviews and summaries of fraudulent activity can serve as institution-based structural assurances—trust-building mechanisms to inform users of a review platform of a machinery at the back end that filters out fraudulent reviews from the non-fraudulent reviews. Indeed, studies in the trust literature demonstrate that building such structural assurances on a website improves trust (Gefen 1997). Interestingly, although informing users about the mere presence of such filtering mechanisms (the informed silence scenario) performed better than the currently popular silent approach, it performed worse than any other form of disclosing censored fraudulent review information.

It is also interesting to note that, between having fraudulent review information displayed and a summarized heuristic, our subjects displayed a higher trust for the latter. This shows that while users prefer to have structural assurances of the honesty of reviews, they also prefer a structural assurance mechanism that decreases their cognitive burden in processing fraudulent review information, and in turn lead to better decisions. This is validated by behavioral theory on information processing that shows how simplified choice mechanisms in the form of decision support systems and heuristics improve the quality of decision making and help consumers to adapt to complex decision-making situations (Johnson and Payne 1985, Payne et al. 1988, Todd and Benbasat 2000). This theory is further validated in the economics literature in Luca and Smith (2013), where the authors show that readers respond more to the overall evaluation of each college compared with other metrics such as raw information on faculty-to-student ratio, alumni donations, etc., when analyzing the *U.S. News & World Report* college rankings.

4.5. Displaying Fraudulent Review Information as a Deterrence Mechanism

There are other interesting implications to displaying fraud review information. First, it increases the cost to commit fraud by creating a system of reputation. The fraud review literature in economics and marketing (DellaVigna and La Ferrara 2010, Mayzlin et al. 2014) demonstrates how systems enforcing rules serve as significant deterrents to committing future fraud. The current practice of silently deleting fraudulent reviews after detection does not deter dishonest businesses from writing fraudulent reviews, as there are no reputational costs to being caught. In fact, results from our experiment show how this policy is the *least* trustworthy among all the other designs that display fraud. This policy of silently censoring fraudulent reviews has very little consequence for bad actors, and thus fosters an environment where actors may commit more fraud, particularly because the reward for getting past the fraud detection mechanism is high.

Having a trust score and displaying fraudulent reviews not only serve as structural assurances, but may signal that rules are enforced on the platform. In the trust literature, this forms the basis of a trust forming process called *calculative-based trust*. Such a mechanism that enforces rules makes users believe that businesses, as rational actors and acting on their self-interest, are not likely to commit fraud because the cost and penalty of getting caught is high (Doney and Cannon 1997). According to theoretical literature on trust, calculative-based trust predicts that even if the user does not necessarily trust each business on

the platform, or have any prior impressions about the reputation of the platform, they understand that the business will not engage in dishonest behavior for fear of the consequences if caught. We propose that these consequences include getting fraudulent review information on record, which will, in-turn, be relayed to current and future users in the form of detected fraud reviews and reduced trust scores.

To validate our theory of how displaying fraudulent review information can influence the trust-formation process, we ran a supplemental survey on AMT with 191 respondents. Through this survey, we hope to understand consumers' perceptions of fraudulent reviews, accountability, and deterrence. Details on this supplemental survey are presented Section B.4 of Online Appendix B. Notably, we find that 71% of the respondents agree that displaying fraudulent reviews along with the summarized heuristic will deter businesses from committing fraud. In contrast, only 26% of consumers agree that not displaying fraudulent reviews (i.e., the silent approach in our experiment) will stop businesses from committing fraud.

In the same supplemental survey, we also collected consumers' beliefs on fraudulent review information, trust on platforms, and transparency. These results show that 91% of users believe that review platforms should inform users about sellers' fraudulent review activity. Importantly, 80% of users agree that they will trust a review platform more if it displays fraudulent reviews because they believe that businesses are less likely to commit review fraud, which is consistent with our experiments.

In addition, 85% of our respondents agree on transparency—the choice to see fake review information—and agree it is up to the users to decide whether they want to use the fake review information to determine the quality and the integrity of the business being reviewed. This naturally leads us to our third question: *When* do users avail themselves of the choice to view fraudulent information?

Thus far, our results from Experiment I show that there are significant changes to users' behavior on a platform and choice when fraudulent reviews are displayed. Experiment II demonstrates that displaying both fraudulent reviews and a decision heuristic increases the trust that the consumers place on the platform. The results also demonstrate the underlying mechanism behind the trust-formation process. Managers can in turn leverage these results to design platforms that signal proactive and consistent enforcement of *deterrence-based* policies in every interaction that their users have with the platforms. Although we observe that consumers display an implicit preference to view fraudulent reviews before they make a purchase decision from the first and second experiments, we do

not know when these fraudulent reviews serve as a salient source of secondary information to change users' prior beliefs of product quality. Do the valence and content of the fraudulent review matter? To answer these questions, we conducted a third experiment, which we present in the next section.

5. Experiment III—Heterogeneity in the Impact of Displaying Fraudulent Reviews

The main goal of this experiment is to understand *when* users will utilize fraudulent review information when presented with this information. Although the results from the previous experiments demonstrate changes in consumer behavior and an increase in the trust that consumers place on the platform, it is not clear under what initial conditions users are more likely to be affected or if the content of the fraudulent reviews mattered when making a final choice. Specifically, once a review is determined to be fraudulent by a consumer, do consumers infer the self-promotional or malevolent nature of fraudulent reviews? This is particularly important because studies show that consumers are disproportionately affected by negative reviews than by positive reviews (Chevalier and Mayzlin 2006). Thus, firms that are vulnerable to being at the receiving end of a fake negative review campaign¹³ might be affected in a different manner if consumers do not factor in the negative fraudulent reviews appropriately.

In Experiment III, we study the heterogeneous reactions of users to exposure versus censorship of fraudulent reviews. In particular, following the IS literature on trust, we focus on an important dimension of consumer decision making for product evaluation on websites—uncertainty in product quality. Many studies in the IS literature have demonstrated the importance of reducing perceived risk and social uncertainty, especially in online settings (Reichheld and Schefter 2000), and the need for websites to build structural assurances to reduce perceived risk and uncertainty (Jarvenpaa et al. 1999, McKnight et al. 2000). However, almost all of these studies have analyzed product evaluation in the context of e-commerce, where there is uncertainty in consumers' interactions with online vendors. There is little research into consumers' trust and initial product evaluation in the presence of user-generated product recommendation mechanisms where there is a high incentive to encounter fraudulent information.

5.1. Randomized Experimental Design

We tested the impact of fraudulent reviews on subjects' behavior under three different settings of product quality—high, medium, and low—by varying the levels of quality of service in the non-fraudulent reviews.

We also randomized the kind of fraudulent reviews (e.g., negative, positive, mixed) that each subject was exposed to.

We ran the experiment on AMT with 289 subjects from the United States, and we paid our subjects a small participation fee of \$0.50. We designed the experiment in such a way that each user was randomly exposed to a restaurant with a low, medium, or high rating. In the instructions, we informed the subjects that they would be provided with \$2 worth of virtual chips with which to bet on a restaurant, thus making the experiment incentive compatible. Therefore, each subject could potentially win \$2 from the experiment, which is approximately a twofold increase over the standard wage of \$0.40 (Ipeirotis 2010) for a five-minute experiment.

Note that in Experiment III, we used the betting framework from Experiment II, but with a significant difference in the design to suit the goal of the experiment. In Experiment II, we were measuring the trustworthiness of website designs based on how they chose to display fraudulent reviews. Therefore, in Experiment II, each subject could split their bets between two different designs shown to them, depending on the amount of trust they place in the designs.

In Experiment III, our goal is to measure the initial perception of quality for a specific business and then measure how consumers' initial perception of the business changes after being exposed to different kinds of fraudulent reviews (positive, negative, or a combination of both). We instructed the subjects that they would be shown one restaurant's review page and that this restaurant had been entered in competitions in the past.¹⁴ We informed the subjects that we had an actual measure of whether the consumers of that restaurant would find its service to be good or bad based on the results of these competitions. Thus, we explained that the restaurant shown to them could "win" or "lose," and the review page would provide them with all the information that they would need to determine whether the restaurant was likely to win or lose.

We then instructed the subjects that if they bet a certain amount of their \$2 bet *for* the restaurant, and if their answer matched our measurement based on the restaurant's true performance, they would win the amount of money they bet. This effectively meant that the remaining amount in their \$2 worth of virtual chips was bet *against* the particular restaurant and they would win this amount if the restaurant did not perform well based on an actual measure of service. Not betting was not an option. We strongly encouraged the subjects that their bet should be based on their likelihood of visiting or not visiting the restaurant if they were making the same decision in real

life. In this design, we are eliciting the subjects' beliefs about the quality of the restaurant based on how they split their bets for and against the restaurant. Note that subjects could also safely split their bets equally for and against the restaurant if they wanted a guaranteed payment of \$1.

The screenshot of the instructions is presented in Figure 11 and 12 in Online Appendix C.

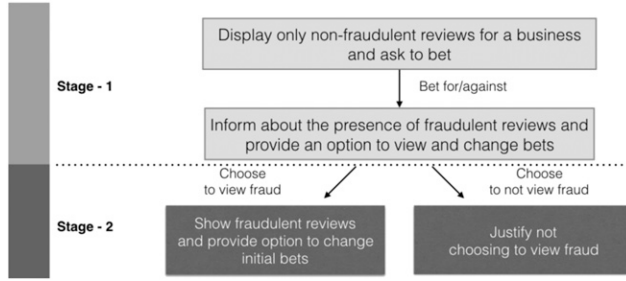
After the instructions, the subjects were randomly shown either a low-, medium-, or high-rated review page of a restaurant. They were presented information usually available on major review portals that present the non-fraudulent reviews after silently deleting fraudulent reviews. (We provide the screenshot of the two stages in Figures 13 and 14 in Online Appendix C.) In the first stage of the experiment, the subjects were asked to bet \$2 for or against the restaurant shown to them based on all the information available to them on the web page of the restaurant, including restaurant reviews. We ensured randomization by performing balance tests. The results are shown in Table 12 in Online Appendix C, Section C.5.

Once they placed their bets based on their initial evaluation of quality or intention to use the service, we disclosed that the restaurant had fraudulent reviews that the review portals had hidden from their view. In this second stage, we provided an option for the subjects to view the fraudulent reviews if they wanted to, and also informed them that they could change their bets after viewing the fraudulent reviews. If the subjects opted to view the fraudulent reviews, they were shown the same review page that they were initially exposed to, except now this review page contained non-fraudulent reviews on the top portion of the page along with distinctly marked fraudulent reviews at the bottom of the page.

At this second stage, we randomized the kind of fraudulent reviews the subjects were exposed to, that is, they could be assigned to a scenario where all the fraudulent reviews were positive (implying self-promotion), negative (implying maligning by competitors), or a mixture of both. If the subjects did not want to view fraudulent reviews, we asked them for an explanation. We did this to enable us to understand whether the subjects trusted the review portal's evaluation of fraudulent reviews. The flow of the experiment is shown in Figure 2.

In the final stage, we asked subjects a set of demographic and internet usage questions. In addition to the monetary incentive, we also added a set of test questions to ensure that the subjects were paying attention.

The aim of the experiment is to see how the subjects' evaluation of product quality changes under different initial conditions with the introduction of fraudulent reviews. Therefore, even if our subjects were trying to

Figure 2. Flow of Randomized Experiment III

guess the business that matched our definition of true quality and performance, we are interested only in measuring how their evaluation changes before and after the exposure to fraudulent reviews, given a particular quality setting of a business.

We first present the number of subjects who chose to view fraud, and the number of subjects who changed their initial bets after viewing fraud, under each quality condition in Table 9. Whereas the percentage of subjects choosing to view fraud remains somewhat similar (with the high condition being the lowest and low condition being the highest), there is significant variation in the number of subjects who changed their bets after viewing the fraudulent reviews. Note that we also randomized the kind of fraudulent review the subjects were exposed to at this stage, when they choose to view fraud (positive, negative, or neutral fraudulent reviews).

To understand the impact of initial perception of quality and the change in perception of quality after revealing fraudulent reviews, which are also of different quality, we analyzed the following model:

$$\begin{aligned}
 \text{ChangeBets}_i &= \beta_0 + \sum_{c=1}^3 \beta_c \text{Quality}_c + \gamma \text{FreqReviewReader}_i \\
 &+ \sum_{j=1}^3 v_j \text{FraudReviewQuality}_j + \sum_{k=1}^3 \alpha_{ik} X_k + \epsilon_i, \quad (4)
 \end{aligned}$$

where *ChangeBets* refers to changes in bets, in cents, after the subjects opted to view fraudulent reviews. This is the difference between the initial bet and the bets placed after viewing the fraudulent reviews. The term *Quality* refers to the quality (high, low, or medium) of the restaurant. The term *FraudReviewQuality* refers to the quality of fraudulent reviews displayed to the subjects, and $j \in \{\text{FraudPositive}, \text{FraudNegative}, \text{FraudMedium}\}$. The term *FreqReviewReader* refers to subjects who read reviews on Yelp/Zagat/Google at least few times a month. The term X_k refers to a vector of demographic controls for the subjects (age, gender, income, and education). We present the summary statistics of the bets placed by the quality condition in Table 10.

Table 9. Viewing and Bet-Changing Behavior of Subjects

Quality	% subjects who chose to view fraud	% subjects who chose to change bets if viewed fraud
High	0.51	0.28
Low	0.65	0.28
Medium	0.53	0.47
ANOVA	$p > 0.05$	$p < 0.05$

Note. ANOVA, Analysis of variance.

5.2. Results

We ran the above specification with 164 subjects who viewed fraudulent reviews across all the quality levels. We present the results of the specification in Table 11.

First, we identify *when* a majority of subjects opt to change their bets after viewing fraudulent reviews. Specifically, we examine the conditions of perceived quality under which subjects used the additional information provided by fraudulent reviews to change their initial evaluations of the business. From Table 11, we observe that the effect is positive and significant for medium-quality restaurants. In other words, we see that fraudulent reviews are likely to overwhelm the subjects' prior impressions of a restaurant of medium quality. Compared with the base of high-quality restaurants, subjects were likely to increase their bets by 18.98 cents for restaurants of medium quality.

Second, we identify whether the valence of the fraudulent reviews made any difference to the change in bets that the subjects placed on a restaurant. We observe the coefficients on the negative or positive fraudulent review indicators are statistically insignificant, which indicates that the valence of the fraudulent review did not matter while changing bets.

5.3. Discussion

5.3.1. Role of Fraudulent Reviews Under Different Prior Evaluations of Uncertainty. The primary role of customer reviews is to help users decrease uncertainty when evaluating the quality of a product using market-generated feedback mechanisms. In this experiment, we study whether uncertainty in product quality can potentially induce users to factor fraudulent reviews in

Table 10. Summary Statistics of Bets Placed by Quality Condition

	Mean	Median	Std. dev.	Min	Max
High	−5.38	0	33.1	−100	100
Low	−5.59	0	34.2	−170	50
Medium	11.4	0	28.8	−50	170

Note. Std. dev., Standard deviation.

Table 11. Impact of Displaying Fraudulent Reviews on Initial Product Quality

	(1) <i>ChangeBets</i> (cents)
Low	1.535 (7.379)
Medium	18.03** (8.258)
<i>FreqReviewReader</i> = 1	−1.658 (28.45)
<i>FraudNegative</i>	3.059 (6.065)
<i>FraudPositive</i>	8.971 (7.391)
Constant	−4.912 (31.07)
<i>N</i>	164

Notes. Control variables are age, gender, income, and education. Robust standard errors are in parentheses.

** $p < 0.05$.

their decision-making process. Note that users do not consider the possibility of fraudulent reviews until the second stage, where we tell them that the restaurant has fraudulent reviews that have been censored. We find that when users were exposed to both fraudulent and non-fraudulent reviews in the second stage (given that they choose to view fraudulent review information), they treated fraudulent reviews as an important supplemental source of information for decision making under uncertainty.

In particular, in the case of decision making for restaurant choice, we observe that more than 50% of our subjects chose to view the fraudulent reviews for restaurants of all qualities, though there is no statistical difference between the quality levels. However, based on Table 9, we observe that 48% of the subjects changed their initial bets on medium-rated restaurants, compared with only 30% for high-rated restaurants and 27% for low-rated restaurants. Therefore, our results suggest that exposing subjects to the possibility of potentially fraudulent reviews induces them to reevaluate their decisions if they had already been uncertain about the quality of the restaurant and choose to seek more information about the restaurant by looking through the fraudulent review information. The uncertainty stems from the fact that medium-rated restaurants tend to demonstrate higher uncertainty in quality (e.g., they tend to have a relatively equal number of positive and negative reviews when fraudulent reviews are not displayed). Notably, after viewing fraudulent reviews, a significantly greater change in bets occurs in the case of medium-rated restaurants, as seen in Table 11. This finding seems to indicate that the impact of fraudulent reviews on final decisions is the highest when the uncertainty in the initial evaluation of the product quality is high, particularly among users who choose to view fraudulent reviews.

In summary, our results demonstrate that subjects who likely had doubts about a restaurant's initial product quality because of mixed reviews and chose to view fraudulent reviews were most affected by viewing fraudulent reviews. This result is also

consistent with the findings in sociological and trust literature that uncertainty causes people to seek assurance in the form of new information, hoping that the new information will help them optimize their decision making (Tversky and Shafir 1992, McKnight et al. 2002)

5.3.2. Valence of Fraudulent Reviews. In the experiment, we randomized the kind of fraudulent reviews that users were exposed to when they opted to view fraudulent review information (i.e., users were exposed to positive fraudulent reviews, which denote self-promotion by a business; negative fraudulent reviews, from competitors who intend to bad-mouth a business; or an equal mixture of both positive and negative fraudulent reviews). We observe potentially significant inefficiency in the users' ability to process fraudulent information when this information is displayed to them.

Our results suggest that although a majority of users in all the conditions viewed fraudulent reviews, and users in the medium-quality condition utilized the fraudulent review information to influence their initial impression of product quality, users were not able to effectively distinguish between different types of fraudulent information when they did choose to use this information. In Table 11, we see that the content (positive, negative, or mixed) of the fraud review was statistically insignificant in determining the overall change in user bets. Intuitively, we would expect users to account for self-promotion and decrease their bets for restaurants with positive fraudulent reviews (by associating positive fraudulent reviews with lower quality). This difference between expected and actual behavior is even more pronounced when we observe the actual change in bets across different fraudulent review conditions. For example, the medium-quality scenario illustrates both rational and irrational behavior on the users' part—58% of the users increase their bets after reading positive fraudulent reviews, whereas we would expect rational customers to decrease their bets to account for self-promotion. However, 40% of the users also increase their bets after reading negative fraudulent reviews, correctly adjusting for maligning competitors.

This kind of error in judgment due to a non-negligible cognitive cost makes a case for the review portals to provide an efficient summary of fraudulent reviews in the form of a simple heuristic such as the trust score proposed in Experiment II. Reducing consumers' mental effort is also particularly important because the results from Experiment III suggests that consumers are more likely to use fraudulent review information when product quality is uncertain. Studies have repeatedly shown that humans are prone to biases when estimating the degree of uncertainty in a situation, and lack consistency when

making intuitive predictions because they do not systematically use all the information presented to them (Kahneman 1982).¹⁵ This is unsurprising given how studies have shown that consumers are likely to accept information in the format presented to them and seldom exert the cognitive effort necessary to process this information (Slovic 1972).

The willingness of firms to decrease prepurchase information scarcity by being more transparent, especially in the realm of experience goods, can signal the efficiency of the review portal in reducing information asymmetry and help to increase consumers' efficiency in making decisions. The proposed trust score not only improves the consumers' decision-making process and trust, but can also be used as a first step in establishing a system of accountability. We discuss an approach to design such a trust score in Online Appendix D. Our example aims to pave a path for future research to help review portals improve information efficiency and user trust.

6. Conclusions and Implications

Consumers place significant weight on the content of online reviews when making purchase decisions (e.g., Chevalier and Mayzlin 2006, Dellarocas et al. 2007, Forman et al. 2008, Chintagunta et al. 2010, Ghose et al. 2012). Fraudulent reviews are an inevitable consequence of this reliance (Smith and Anderson 2016). Recognizing the potential of fraudulent reviews to distort consumer behavior and potentially weaken consumer trust, firms invest significant engineering and managerial effort to identify potentially fraudulent reviews. For example, Yelp has identified 16% of the total reviews on its platform as fraudulent (Luca and Zervas 2016). Similarly, in 2015, using detailed user and behavioral patterns, TripAdvisor identified 60 offshore review farms (Parkin 2018), and Amazon sued 1,000 people for writing fraudulent reviews on its platform (Conger 2016).

However, although there has been much research in the information systems and computer science literature studying how to algorithmically detect fraudulent reviews in online platforms, ours is the first study that we are aware of to analyze the equally important question: What should firms do *after* they discover potentially fraudulent reviews. This question has important academic and practical implications given the lack of consensus on what to do with potentially fraudulent reviews. For example, in the instances mentioned above, Amazon chose to silently censor potentially fraudulent reviews, whereas Yelp was much more public about the existence of fraudulent reviews: identifying fraudulent reviews and moving them from the main review section to a "non-recommended" section.

Our research seeks to inform both the academic and managerial understanding of this question using a variety of experimental techniques. Our results have several practical implications for platform managers. First, our results show that platforms can increase consumer trust by leaving potentially fraudulent reviews on their sites instead of silently censoring these reviews. Second, and critically, our results provide guidance to managers about how potentially fraudulent reviews should be displayed to users. Specifically, our results show that fraudulent review information is best displayed using a decision heuristic that decreases the users' cognitive burden in reconciling the overall informational signal derived from genuine reviews with fraudulent reviews. Third, we find that although there are important motivational differences between positive fraudulent reviews and negative fraudulent reviews (Luca and Zervas 2016), consumers are not effective at processing this information. Fourth, our results suggest that the transparent strategy of managing fraud is effective on the demand side because it increases consumers' overall trust in the reliability of reviews. Said another way, our results show that any decrease in trust a platform may face from admitting to naive users that there is fraud on their site is dominated by the increase in trust from consumers who already believed that there was fraud and now know that something is being done to alleviate it. Fifth, our results suggest that a transparency strategy is effective on the supply side by providing appropriate disincentives to self-interested parties who may consider posting fraudulent reviews in the first place.

In addition to influencing managerial practice, our results highlighting the importance of transparency over censorship may have implications for public policy. Just as there are strong incentives to fraudulently manipulate consumer beliefs pertaining to commerce, there are also strong incentives to fraudulently manipulate individual beliefs pertaining to public policy decisions. For example, there is a growing interest from legislative bodies and enforcement agencies around the world into how fake information that can distort consumers trust in online markets (Bhattacharya 2017). Likewise, policy makers have increased their focus recently on the impact of "fake news" and other fraudulent information on voter behavior. Our research could be extended to inform both sets of policy questions by providing appropriate disincentives to suppliers of fake information and also providing appropriate confidence to consumers in the reliability of platforms of online information.

Nonetheless, we note that our analysis has several limitations. First, our experiments were conducted using AMT. Although the behavioral research literature (Paolacci et al. 2010, Suri and Watts 2011,

Parkes et al. 2012) has consistently shown that experiments conducted on AMT provide results equivalent to those of experiments conducted in a laboratory, ideally, we would have preferred to observe real users in a real-world setting. Although randomization of our experiments helps to reduce the concern of heterogeneity and selection, as in any experimental work, it is difficult to evaluate how users would behave under real-world conditions and what they would perceive as the inherent risk in making a bad decision while looking at reviews. Second, we acknowledge that trust is not a one-dimensional quantity, and in our experiment, we considered only a setting where the reviewer had no prior knowledge about the portal.

We also acknowledge that there might be other practical considerations that prevent platforms from displaying fraudulent reviews. First, platforms might not want to draw attention to the fact that there are fake reviews on the platform. This approach can indicate the presence of fraudulent reviews on the platform and can negatively impact the consumers' trust on all the reviews on the platform. However, the strategy of silently censoring fraudulent reviews has caused a negative backlash in the media, particularly over the lack of transparency (Barnett 2015). The sporadic legal action by Amazon can impact specific sellers' behavior; however, not acknowledging the problem or taking a public stand against it can exacerbate consumers' mistrust of technology firms (Mitchell et al. 2019).

Second, there is a conflict of interest between businesses that advertise on tech platforms and are also reviewed on the same platforms. Platforms could be viewed as suppressing positive reviews as fraudulent if a business refuses to advertise with the platform and the public is aware that such a system is in place. Yelp has been sued (unsuccessfully) by businesses accusing the firm of displaying negative reviews as way to extort businesses' into advertising, and has been the subject of many related conspiracy theories regarding their behavior in censoring reviews. However, tech firms that do not censor fraudulent reviews have also been sued (unsuccessfully) by businesses over negative reviews (Roberts 2011). In fact, by displaying the "non-recommended reviews," Yelp has been able to dispel the myths surrounding their conflicts of interest.¹⁶

Finally, firms might not want to display fraudulent reviews because of the fear of fraudsters reverse engineering the fraud detection algorithm. However, reverse engineering fraud detection algorithms is difficult because machine learning-based algorithms factor thousands of user-facing parameters and change dynamically to cope with newer and more innovative forms of fraud.

In spite of these limitations, we believe that our paper advances the academic understanding of how consumers respond to fraudulent information online, advances the state-of-the-art practice in the industry for dealing with the presence of fraudulent reviews, advances future research by providing a more detailed understanding of user behavior and review dynamics inherent to online fraudulent information, and may inform future regulatory and policy discussions of the widespread incidence of fake information being disseminated online.

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Endnotes

¹ See, for example, Jindal and Liu (2008) and Ott et al. (2011), who show that fake reviews can be identified with high degree of accuracy by using a combination of text mining and natural language processing, and Li et al. (2014, 2017), Mukherjee et al. (2012), Xu and Zhang (2015), and Banerjee et al. (2014), who use behavioral methods such as group spam identification, rating review posting, geo-locations, internet protocol addresses, and keystrokes to identify potentially fake reviews.

² See Google Local's review policies (<https://support.google.com/business/answer/2622994?hl=en>).

³ See <https://stories.flipkart.com/flipkart-fake-reviews/>.

⁴ Yelp's policy is to provide the number of the suspected fake reviews on the profile page of each business. The text for these suspected reviews is made accessible to customers from a separate page.

⁵ Cities were added in our experiment to remove the possibility of city-level differences potentially influencing the results. The cities used in our study were Atlanta, Austin, Los Angeles, New York, and Pittsburgh. The subjects were each randomly assigned to a main page with a collection of 50 restaurants from one of these five cities.

⁶ The restaurant data were taken from Yelp.com by using the first 50 results in the search for the term "Restaurants" in each of the five cities used in our study. Thus, the selection of restaurants displayed to the users is a representative sample of restaurants from Yelp in each of the cities.

⁷ For example, Smith and Walker (1993) show how increasing monetary rewards shift the central tendency of the data toward the predictions of rational models.

⁸ Note that this represents the restaurant page after the initial list of restaurants shown in Step 2 of Figure 3 in Online Appendix A.

⁹ Recent news articles show how fake reviews have increased not only in platforms like Amazon, but also on other platforms such as those of Walmart and Sephora (Morris 2019).

¹⁰ See <https://www.tripadvisor.com/hc/en-us/articles/200614797-Our-guidelines-for-traveler-reviews>.

¹¹ Individual restaurant review pages inside each design of the review portal were the same and differed only in the way fraudulent review information was displayed in the corresponding scenario.

¹²We obtained the restaurant data from Yelp. We retained the main template of Yelp, but we removed the platform name and changed the design depending on the scenarios that we wanted to test in this experiment. Please refer to Figure 1 for the changes in design.

¹³Negative fraudulent reviews are as common as positive fraudulent reviews. In fact, spammer communities that write fake reviews are known to convince competitors to write negative fraudulent reviews if the focal business in the same geographical location has already started soliciting self-promotional positive fake reviews (Li et al. 2017)

¹⁴We obtained the restaurant data from Yelp. We changed the quality of the restaurant shown to the users by showing a majority of positive reviews (to indicate high quality), negative reviews (to indicate low quality), or a mixture of positive and negative reviews (to indicate medium quality) and adjusting the overall rating according to the quality setting

¹⁵Although Experiment I shows that users chose restaurants with lower average fraud ratios, note that users viewed only the total number of fraudulent reviews in Experiment I, that is, when the restaurant page was open and they did not know the valence of the fraudulent reviews from the restaurant's main page.

¹⁶See https://www.yelp.com/advertiser_faq.

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